

Task 1: Condition classifier

The treatment condition of cells was determined from their gene expression profile using a gradient boosted tree (xgboost) and a multilayer perceptron (MLP). To evaluate the classification, 5-fold CV was used. The MLP achieved a balanced accuracy of 0.9850 ± 0.0009 across folds. The boosted tree achieved a comparable balanced accuracy with 400 rounds, while a tree with 6 rounds performed weaker. The confusion matrices show that the largest error for each model is cells from IFN treatment being classified as Control cells with around 2% of cells in that error in each model.

Feature importances for MLP and xgb are determined by $|\text{gradient} \times \text{input}|$ and SHAP respectively.

The genes were ranked by feature importance for each classifier and the classifiers were compared by spearman correlation of all genes, Kendall rank correlation coefficient and Jaccard index. The MLP and 400-estimator boosted tree show the highest similarities, however, there are differences in the most important features for classification used by each model. In the top 50 genes, 34 overlap between the MLP and 400-estimator tree.

Marker genes identified in the original paper by Frangieh *et al.* could partly be identified in our analysis. Of the 29 markers from the paper which were explicitly included in the feature-set used for training of the classifiers, 10 were in the top 20 most important features of the tree classifier and 5 were in the top 20 most important features of the MLP.

Biologically, a large number of genes ranking high in the feature ranking are involved in antigen presentation and immune regulation, such as CD7, CD74, CXCL9, CXCL10, GBP1, GBP2, GBP4, GNLY, HLA-B, HLA-DRA, HLA-DRB1, IDO1, LAP3, PTGES, and WARS. This is an expected result since the main difference between the treatment conditions is interferon signalling and interaction with TILs in co-culture conditions. Therefore, observing differences in gene expression affected by immune signalling related pathways is an expected finding.

Task 2: Perturbation clustering

Clustering was performed at the perturbation level. Within each condition, all QC-passing cells with the same value in `obs["perturbation"]` were aggregated by calculating their mean expression profile. Perturbations represented by fewer than 50 cells were excluded. This retained 221 perturbations in co-culture, 219 in control and 229 under IFN- γ stimulation. The selected genes already reflected the expected condition-specific biology: antigen-presentation genes such as CD74, HLA-DRA, HLA-DRB1 and HLA-DPA1 had the highest excess variance in co-culture, whereas GBP1, GBP2, GBP4, IDO1 and WARS dominated the IFN- γ condition.

The clearest and most reproducible group was the IFN- γ /JAK-STAT module. In co-culture, DBSCAN, Ward clustering and k-means all placed IFNGR1, IFNGR2, JAK1, JAK2 and STAT1 in the same cluster. HDBSCAN recovered these five genes together with CDKN2B and SNHG6. Under IFN- γ stimulation, HDBSCAN produced an equivalent six-member group containing CDKN2B and all five IFN- γ /JAK-STAT genes. Its perturbation profile showed strong reductions in canonical IFN-response genes, including GBP1, GBP2, GBP4, IDO1 and WARS.

The failure to mount the transcriptional response to IFN- γ agrees with the study by Frangieh et al., which described defects in IFN- γ sensing and JAK-STAT signalling as an important mechanism of resistance to T-cell pressure.

The MHC-I and CDK4/6 modules were recovered only weakly. The highest MHC-I Jaccard similarity was 0.133, from control-condition k-means, while the best CDK4/6 similarity was 0.200, from control-condition DBSCAN. In co-culture, their best similarities were only approximately 0.11 and 0.14, respectively. CD58 also did not form an independent group and instead appeared within a larger, heterogeneous HDBSCAN cluster. This is not necessarily inconsistent with the paper. CD58 represents a distinct co-stimulatory mechanism and was reported to act separately from MHC-I expression and IFN- γ signalling. Because it is represented by a single perturbation, conventional cluster-recovery measures may not meaningfully identify a "CD58 module."

For the co-culture condition, Ward clustering had the strongest agreement with the paper modules, with a paper-label ARI of 0.418 and exact recovery of the five-member IFN- γ /JAK-STAT module. However, its silhouette score was negative at -0.035 and its bootstrap stability was only 0.125. HDBSCAN produced more internally separated clusters: its co-culture silhouette score was 0.242, paper-label ARI was 0.360 and stability was 0.201. Its limitation was coverage. DBSCAN recovered the exact five-member IFN- γ /JAK-STAT module in both co-culture and IFN- γ conditions. Nevertheless, it assigned 216 of 221 co-culture perturbations and 224 of 229 IFN- γ perturbations to noise. Its apparent stability of 1.0 therefore reflects repeated recovery of the same very strong five-member group, not stable clustering of the complete dataset.

Across all conditions, IFN- γ HDBSCAN was selected as the best non-degenerate result using the product of paper-label agreement and bootstrap stability. It had a paper-label ARI of 0.330, stability of 0.279 and silhouette score of 0.211. It still classified 174 of 229 perturbations as noise, so this selection should not be interpreted as evidence that HDBSCAN explains the complete structure of the data.

Task 3: Predict effect of unknown perturbation on gene expression

Two models, ridge regression and a neural network, were used to predict the gene expression profile of held-out perturbations and compared to two intentionally simple baselines. The feature set of the neural network included pathway-membership of the perturbed gene in addition to self-expression, co-expression and condition features, while only the latter were selected for training of the regression model. As baselines, a zero-model and mean-model was used, where the zero-baseline predicts a log2FC of zero for all genes for all perturbations and the mean-baseline predicts the mean log2FC for each gene over all training perturbations. The rationale for the zero-baseline is the assumption from biological knowledge that most perturbations have little effect on the expression of most genes. The rationale behind the mean-baseline is the assumption that most perturbations from the screen have similar effects on gene expression.

The results show that the more complex models do not reliably perform better than the baselines. Comparing RMSE (calculated using a random holdout of 10 perturbations from the 50 selected perturbations) of the ridge regression to the mean-baseline reveals a non-significant mean increase of 0.0010 while the RMSE of the neural network also did not significantly differ compared to either the mean-baseline or the ridge regression model.

To test generalization to perturbations in new pathways, IFN- γ and MHC-I modules were held out and performance compared to a random holdout. For whole-module holdouts, Pearson r decreased slightly but not significantly indicating that generalizing to new pathways is a harder task but possible. Confirming the results from task 2, generalizing from perturbations to MHC-I is simpler than to IFN- γ , signified by lower RMSE and higher Pearson r for MHC-I holdout.

Regarding the bias-variance-tradeoff, the baselines are the most biased and least variant, as they assume very specific behaviours of the perturbations (no change in log2FC or the same change in log2FC for all perturbations). Ridge regression has more variance, but a bias is introduced by coefficient shrinking to reduce variance and overfitting. The neural network is even less biased and has more variance, which at the same time increases the risk of overfitting and modelling noise. In this case, the more complex models are not able to significantly improve model performance over the simple baselines.

With the protein expression data, model performance is weak, as the neural network predicts near zero log2FC for all targeted protein markers. While most perturbations do not strongly affect expression of the panel proteins, the model does not generalize well to the perturbations that do.

Since with 50 selected perturbations, only a small amount of training data is available to predict a feature which is highly variable and noisy in biological context, good prediction performance should not be expected. The prediction quality of both models may likely increase when a larger set of perturbations is used for training. However, the difference to the baselines in our examples is so small, it is possible that training on more perturbations might still not significantly improve the more complex models over simple baselines.